MLOPS Assignment 2

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1. Which run had the lowest latency, and why might that be?

Inthefirstexecution,**run_batch4**hadthelowestlatency(0.034s),followedbyrun_batch8(0.047s) and run_batch1 (0.163 s). Processing multiple sentences together reduces repeated inference-call overhead. The exact timings can vary between executions because of warm-up and system conditions.

2. Did accuracy change between runs? Why or why not?

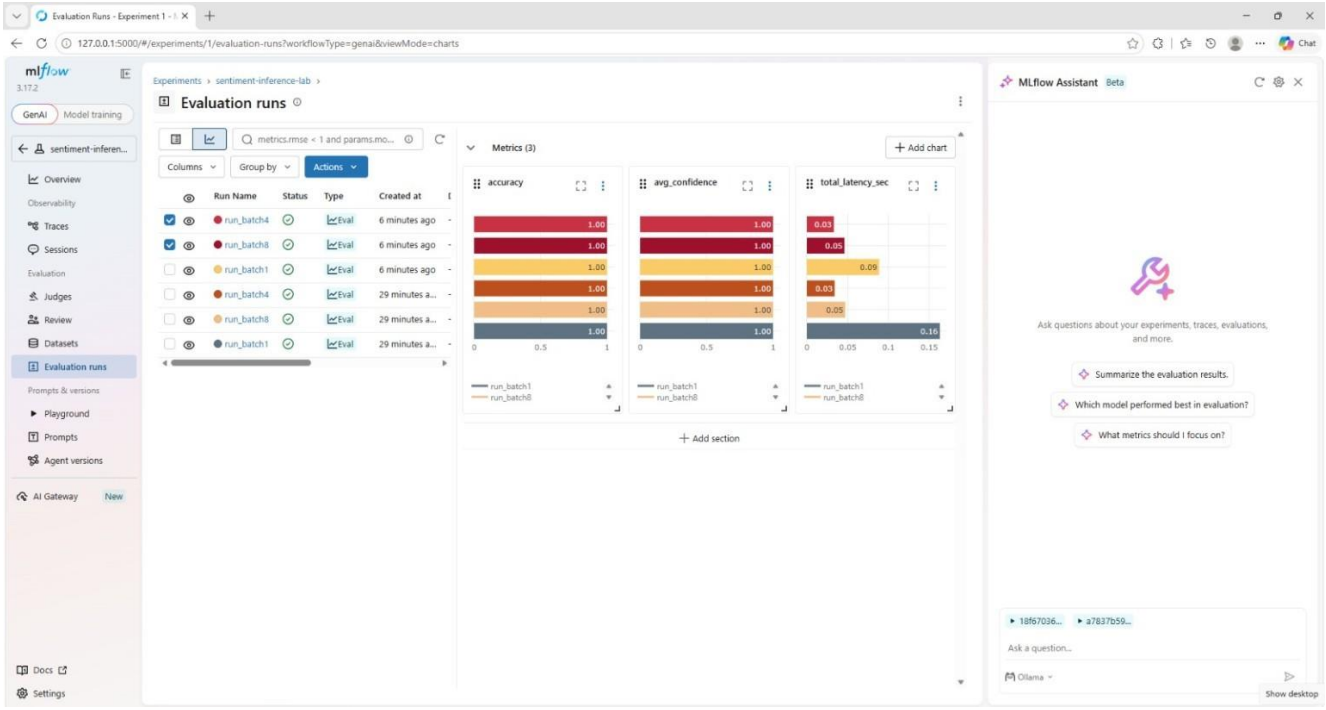
No.Accuracyremained**1.0(100%)**forallthreebatchsizes.Thesamepretrainedsentiment-analysis model and test sentences were used. Changing the batch size affects how inputs are grouped for inference, not the model weights or expected predictions.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

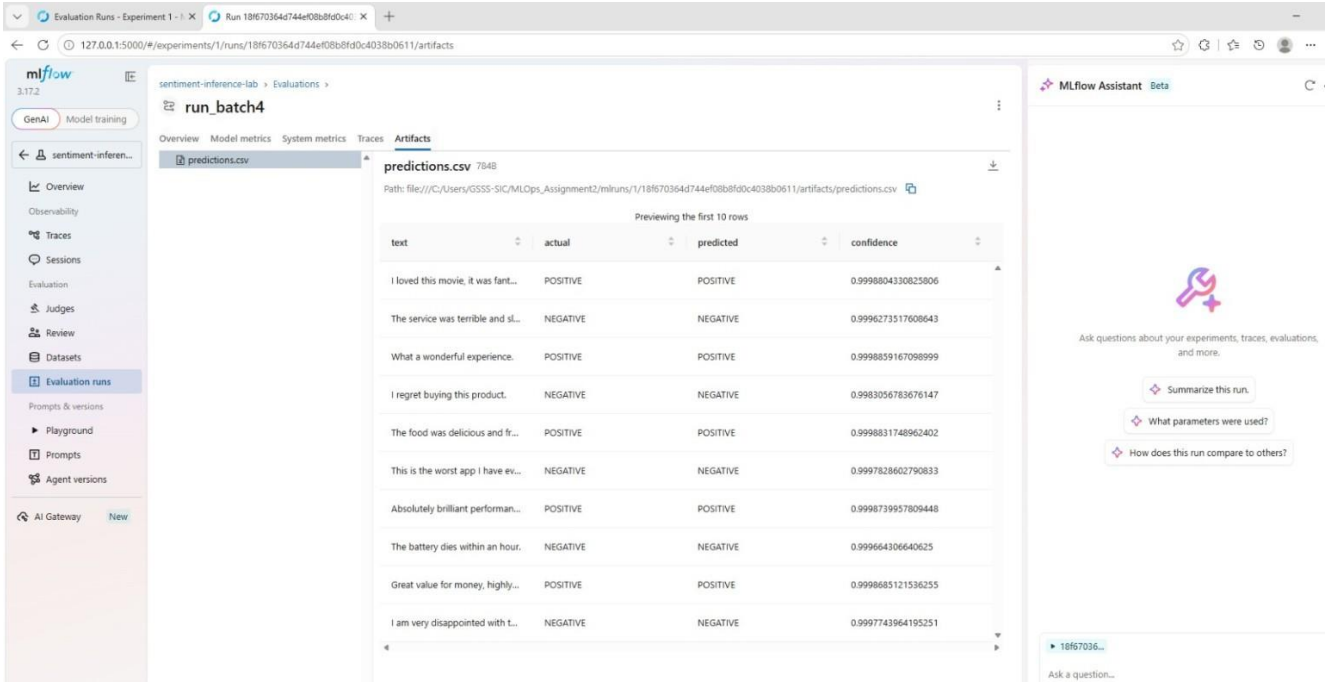
Reproducibilityandrecord-keeping:MLflowsavesparameters,metricsandartifactssuchas predictions.csv for later inspection.

Easycomparison:MLflowdisplaysrunsandtheirmetricssidebyside,makingiteasiertocompare accuracy, confidence and latency than reading terminal output alone.

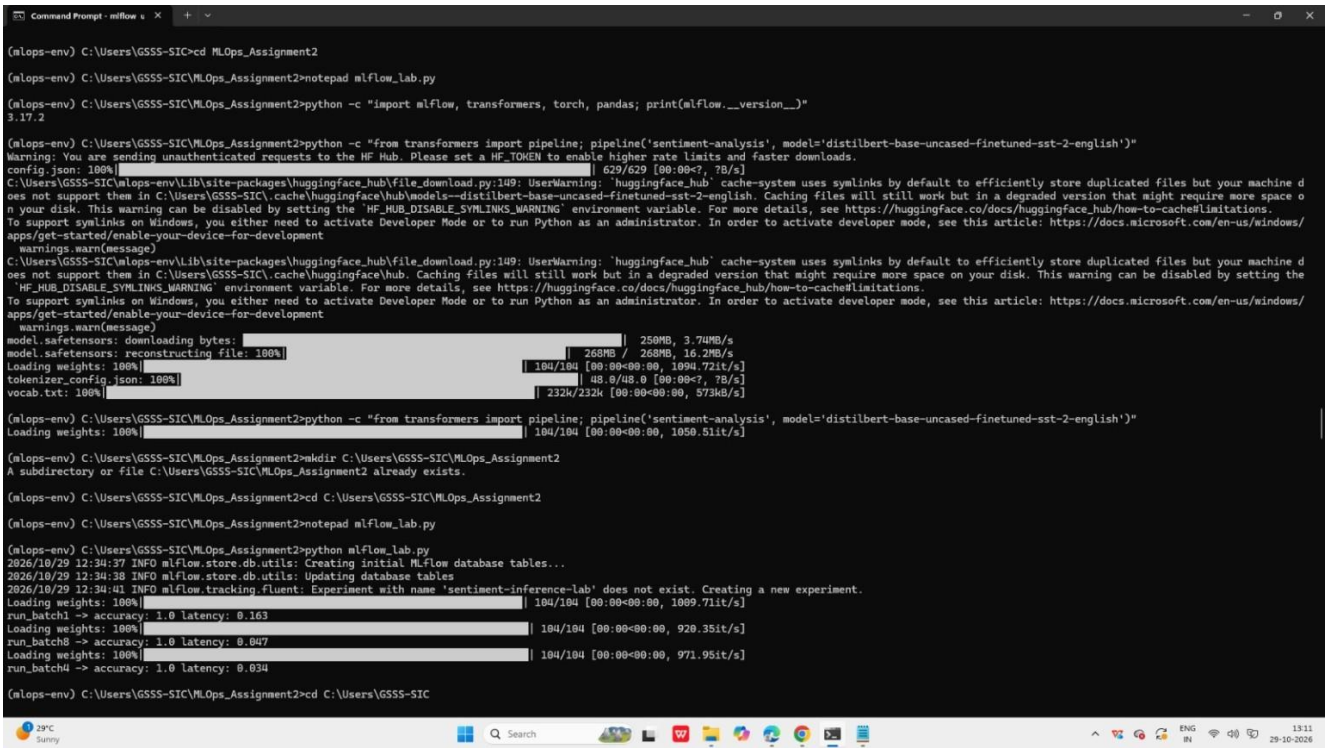
Screenshot of the MLflow runs table and comparison charts:



Screenshot of the run_batch4 artifact (predictions.csv):



Execution: First run and recorded metrics:



Execution: MLflow server and subsequent run:

